

**BRITISH AIRWAYS –
CUSTOMER BOOKING
PREDICTION MODEL**

AGENDA

Objective

Data Preparation

Model Development

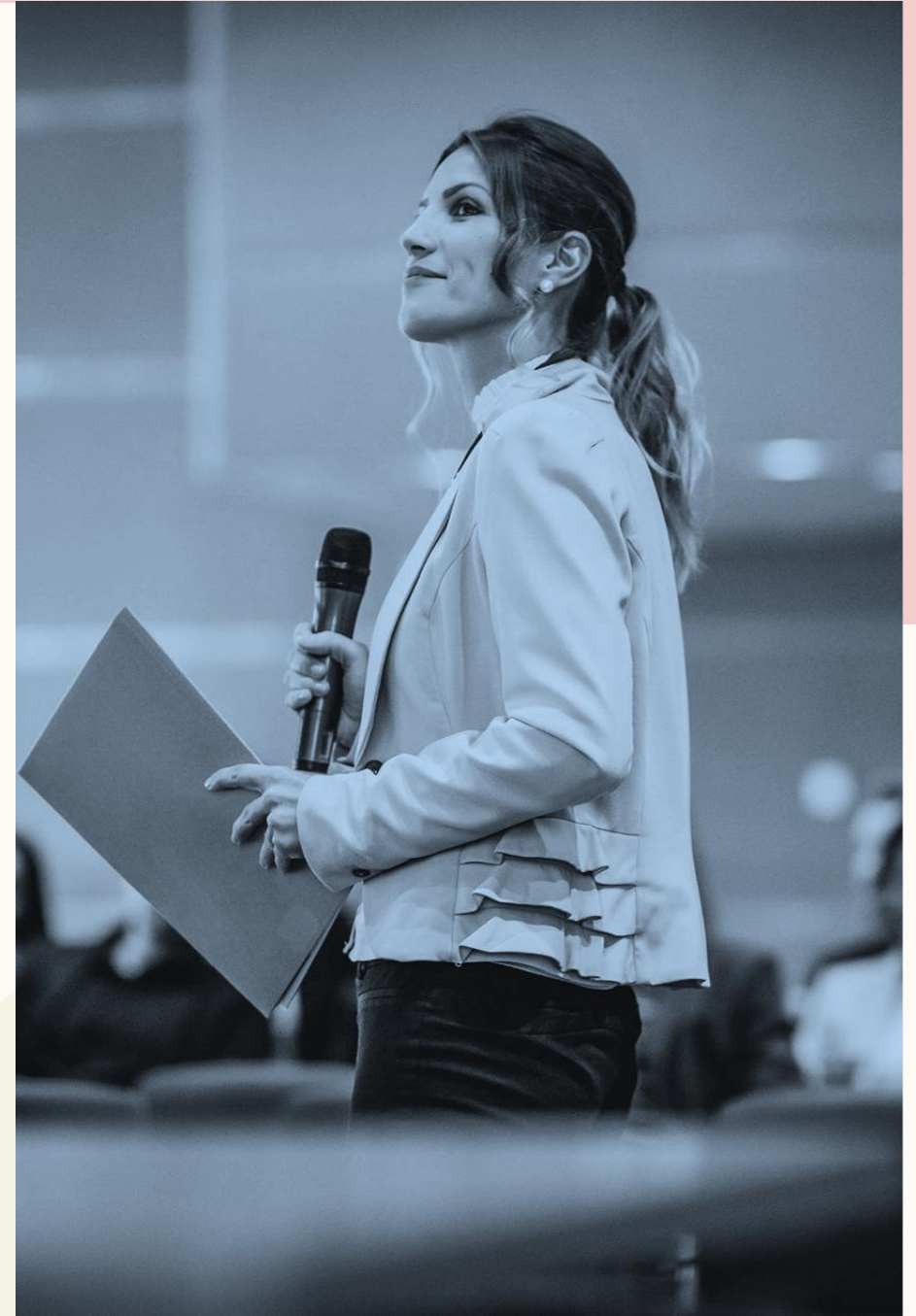
Model Performance

Key Drivers of Booking Completion

Business Insights & Recommendations

OBJECTIVE

To predict customer booking behaviour and understand what factors make customers more likely to complete a booking.



DATA PREPARATION

- Reviewed the customer booking dataset.
- Checked for missing values and data quality issues.
- Converted categorical variables (e.g., sales channel, trip type, route, booking origin) into numerical values using Label Encoding.
- Converted flight day from text format (Mon, Tue, Wed, etc.) into numerical values.
- Separated the data into:
- **Features (X)**: customer and booking information.
- **Target (y)**: booking completion (0 = No Booking, 1 = Booking Completed).
- Split the dataset into training data (80%) and testing data (20%) for model development.

MODEL DEVELOPMENT

- Selected **Random Forest Classifier** as the machine learning algorithm.
- Trained the model using 80% of the dataset.
- Used the remaining 20% of the data for testing.
- Generated predictions on customer booking completion.
- Evaluated model performance using accuracy, classification metrics, and cross-validation.
- Analyzed feature importance to identify the key factors influencing booking behaviour.

MODEL PERFORMANCE

- Random Forest model achieved an **accuracy of 85.41%**.
- Average **cross-validation score: 48.56%**.
- The model performed well on the test dataset but showed lower consistency across different data samples.
- This suggests the model can predict bookings reasonably well, but further tuning could improve reliability.

Key Metrics

Metric	Result
Accuracy	85.41%
Cross-Validation Score	48.56%

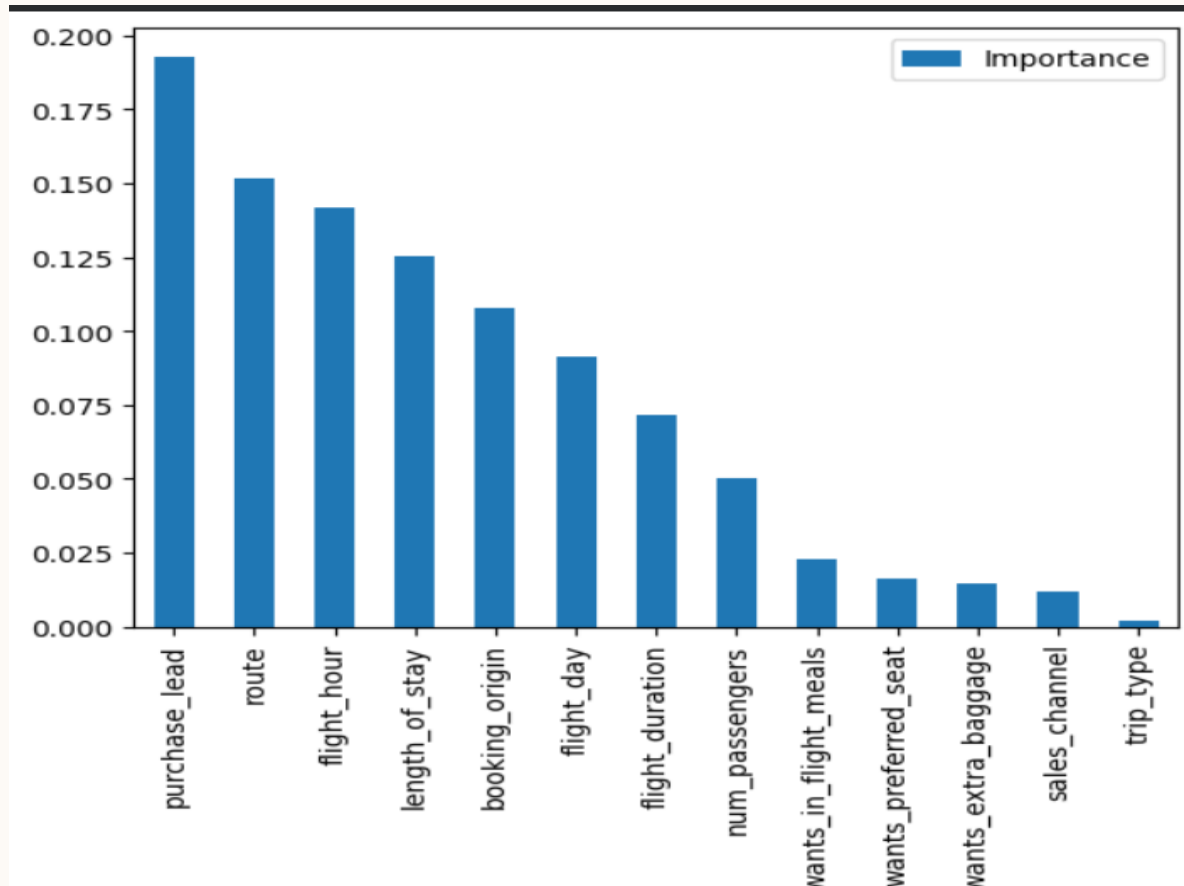


KEY DRIVERS OF BOOKING COMPLETION

- **Purchase Lead** was the most important factor. This means customers who book their flights well in advance are more likely to complete their booking.
- **Route** was the second most important factor. Some flight routes are more popular and have higher booking completion rates than others.
- **Flight Hour** also played an important role. The time of departure can influence a customer's decision to book.
- **Length of Stay** was another key factor. The number of days a customer plans to stay at their destination affects booking behaviour.
- **Booking Origin** was important as well. Customers from different countries may have different booking patterns and preferences.



CHART



BUSINESS INSIGHTS

- Customers who book flights earlier are more likely to complete their booking.
- Flight route and departure time play an important role in customer decisions.
- Customer travel patterns, such as length of stay and booking origin, help predict booking behaviour.
- Additional services (baggage, meals, and preferred seats) have less influence on booking completion.

RECOMMENDATIONS

- Target customers who book well in advance with personalized offers and promotions.
- Focus marketing campaigns on key routes and peak booking periods.
- Use customer booking patterns to improve targeting and increase conversion rates.
- Continue improving the model by adding more customer and booking data to increase prediction accuracy.

**THANK
YOU**